

Problem Set for 08 May 2025: A New Jordan form over $\mathbb{C}$

Important

Throughout, $\mathbb{C}$ is the base field. Define $|z| := \sqrt{z\bar{z}}$ for $z \in \mathbb{C}$.

Definition (Eigen-). Let $A \in \mathbb{C}^{n \times n}$. We say that (v, λ) is a eigenpair of A provided $A\bar{v} = \lambda v$ for some $\lambda \in \mathbb{C}$. The vector v is referred to as a eigenvector and λ a eigenvalue, correspondingly. For matrices, we say A is similar to B provided $A = PBP^{-1}$ for some non-singular matrix P .

Exercise 1 Characterise the eigenspace corresponding to the eigenvalue 0.

Warning

Eigenspaces associated with eigenvalues $\neq 0$ are not, in general, $\mathbb{C}$ -vector spaces.

Exercise 2 Demonstrate that if λ is a eigenvalue, then so too is $e^{i\theta} \cdot \lambda$. Furthermore, show that for any eigenvector v , there exists a eigenvector associated with a eigenvalue ≥ 0 .

Exercise 3 Suppose (v, σ) is a eigenpair of A ; then $(v, \sigma\bar{\sigma})$ is an eigenpair of $A\bar{A}$. In particular,

- if $A \in \mathbb{C}^{d \times d}$ possesses d eigenpairs which are linearly independent over $\mathbb{C}$, then $A\bar{A}$ is diagonalisable with all eigenvalues non-negative.

Now consider the following **partial converse statement**:

- A has no eigenvectors if and only if $A\bar{A}$ admits no eigenvalues in $\mathbb{R}_{\geq 0}$;
- $A \in \mathbb{C}^{d \times d}$ has d linearly independent eigenpairs over $\mathbb{C}$ if and only if $A\bar{A}$ is diagonalisable with non-negative eigenvalues.

Prove the **partial converse statement**: suppose that $(v, \sigma\bar{\sigma})$ is an eigenpair of $A\bar{A}$; then σ is a eigenvalue of A .

Tip

Construct an associated eigenvector of some $e^{i\theta}\sigma$, then invoke previous exercises.

Warning

When A has geometric multiplicity 1, one may simply take the eigenvector $e^{i\theta}v$; however, in general, $e^{i\theta}v$ is not a eigenvector. The construction closely resembles that found in **Exercise 6**.

Exercise 4 Elucidate **Exercise 3** (particularly the warning) with the following example:

$$A = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Exercise 5 This serves to complete **Exercise 2**. Suppose $\{(v_i, \lambda_i)\}_{i=1}^n$ are eigenpairs of A such that $|c_i| \neq |c_j|$; show that $\{v_i\}_{i=1}^n$ are linearly independent.

Exercise 6 We aim to identify the analogue of the Jordan canonical form in the c-version, namely, $X = PJ\bar{P}^{-1}$. We commence with a lemma:

- $P\bar{P}^{-1} = I$ if and only if $P = Q\bar{Q}^{-1}$ for some invertible Q .

 Tip

Take $Q = a\bar{P} + bI$.

Exercise 7 The most elementary Jordan matrix is the diagonal matrix. Show that A is cdiagonalisable (i.e., $A = PD\bar{P}^{-1}$ for some P) if and only if $A\bar{A}$ is diagonalisable. In this scenario, $A\bar{A}$ has only non-negative eigenvalues.

 Note

In this diagonalisable case, one may also verify that $\text{rank}(A) = \text{rank}(A\bar{A})$; that is, the multiplicity of 0 is preserved.

Exercise 8 (Nilpotent part). Show that there exists a matrix P such that $PA\bar{P}^{-1} = \begin{pmatrix} N & O \\ O & B \end{pmatrix}$, where N comprises nilpotent Jordan blocks and B is invertible.

 Tip

Consider the method we used last term to find $J_{s_i}(0)$'s. Useful hints can be found in **Exercise 11**.

Exercise 9 To analyse the Jordan form of A , we begin by studying the Jordan form of $A\bar{A}$. Demonstrate the following:

1. $A\bar{A}$ has a characteristic polynomial with real coefficients;
2. non-real Jordan blocks occur in conjugate pairs;
3. if $\dim \ker(\lambda I - A\bar{A}) = 1$ and $\lambda \in \mathbb{R}$, then $\lambda \geq 0$;
4. **(optional)** Generalise item 3 by showing that for any eigenvalue $\lambda < 0$, the dimension of the null space $\dim \ker(\lambda I - A\bar{A})^p$ is always even ($\forall p \geq 1$), implying that negative Jordan blocks occur in pairs.

 Warning

This is a straightforward exercise. Nonetheless, note that $A\bar{A}$ may possess negative eigenvalues.

Exercise 10 Show that there exists a matrix P such that PAP^{-1} is block diagonal with the following summands:

1. (Jordan type): $J_s(\lambda)$ where $\lambda \geq 0$;
2. (Half-Jordan type): Let $H_{2s}(\lambda) := \begin{pmatrix} O & J_s(\lambda) \\ I & O \end{pmatrix}$.
 1. (For non-real eigenvalues): Since each complex Jordan block in $A\bar{A}$ appears in a pair, i.e., $\begin{pmatrix} J_s(z) & O \\ O & J_s(\bar{z}) \end{pmatrix}$, define $H_{2s}(z)$ as a block of PAP^{-1} satisfying $H_{2s}(z) \cdot H_{2s}(z)^{-1} = \begin{pmatrix} J_s(z) & O \\ O & J_s(\bar{z}) \end{pmatrix}$. $H_{2s}(z) \cdot \overline{H_{2s}(z)} = \begin{pmatrix} J_s(z) & O \\ O & J_s(\bar{z}) \end{pmatrix}$.
 2. (For negative eigenvalues): For a negative eigenvalue λ of A , the Jordan blocks also appear in pairs (see **Exercise 9**).

We refer to this as the cJordan form of A . It is unique.

Tip

The uniqueness of non-zero J - and H -blocks is governed by $A\bar{A}$; the uniqueness of the $J(0)$ -portion is determined by the rank sequence $\text{rank}(A)$, $\text{rank}(A\bar{A})$, $\text{rank}(A\bar{A}A)$, and so forth.

Exercise 11 (optional) (The uniqueness between the cJordan form of A and the Jordan form of $A\bar{A}$). The cJordan form is essentially unique.

- Find distinct matrices $A \neq B$ such that $A\bar{A} = B\bar{B}$;
- Suppose $A\bar{A} = B\bar{B}$ are invertible; then there exists a matrix P such that $PAP^{-1} = B$. In this case, we say A is scimilar to B .

In summary, A is csimilar to B if and only if the following hold:

1. $A\bar{A}$ is similar to $B\bar{B}$;
2. $\text{rank}(A\bar{A}A \cdots A(\bar{A})) \equiv \text{rank}(B\bar{B}B \cdots B(\bar{B}))$; that is, A and B possess the same $J(0)$ -portion in their cJordan forms.

Exercise 12 (optional) Utilise the cJordan form to show that any complex matrix A is csimilar to:

1. λA for $|\lambda| = 1$;
2. $\bar{A}$;

3. A^T ;
4. a real matrix;
5. a Hermitian matrix.

Exercise 13 (optional) Show that for any complex square matrix A ,

$$\det(AA^H + I) \geq \det(A\bar{A} + I) \geq 0.$$

 Tip

Apply the identity $AA^T + BB^T = RR^T$, and the Cauchy–Binet formula to establish the first inequality.

Exercise 14 (optional) Using cJordan forms, show that any complex matrix is the product of a symmetric matrix and a Hermitian matrix.

 Note

It is known that any real or complex matrix can be expressed as the product of two symmetric matrices.

Exercise 15 (optional) Show that $\begin{pmatrix} O & A \\ A & O \end{pmatrix} \sim \begin{pmatrix} O & B \\ B & O \end{pmatrix}$ if and only if $\begin{pmatrix} O & A \\ -A & O \end{pmatrix} \sim \begin{pmatrix} O & B \\ -B & O \end{pmatrix}$.

 Tip

Equivalently, if and only if A and B are csimilar.